

Physical Chemistry Master Formula Sheet

Solutions, Electrochemistry, Kinetics & Thermodynamics

Chapter / Focus Area: Class 11 & 12 Physical Chemistry Formula Handbook

1. Solutions & Colligative Properties

- Raoult's Law: $P_{\text{total}} = p_A^\circ x_A + p_B^\circ x_B$
 - Relative Lowering of Vapour Pressure: $(p^\circ - p)/p^\circ = i \cdot x_{\text{solute}}$
 - Elevation in Boiling Point: $\Delta T_b = i \cdot K_b \cdot m$
 - Depression in Freezing Point: $\Delta T_f = i \cdot K_f \cdot m$
 - Osmotic Pressure: $\pi = i \cdot C \cdot R \cdot T$
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2. Electrochemistry & Chemical Kinetics

- Nernst Equation: $E_{\text{cell}} = E^\circ_{\text{cell}} - (0.0591/n) \cdot \log(Q)$ at 298 K
 - Kohlrausch's Law: $\Lambda^\circ_m = \nu_+ \cdot \lambda^\circ_{+} + \nu_- \cdot \lambda^\circ_{-}$
 - First Order Kinetics: $k = (2.303/t) \cdot \log([A]_0/[A]_t)$, $t_{1/2} = 0.693/k$
 - Arrhenius Equation: $k = A \cdot \exp(-E_a / RT)$, $\log(k_2/k_1) = (E_a/2.303R) \cdot (1/T_1 - 1/T_2)$
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